CHM_159
Worksheet_Chapter_6

Valence Bond Theory, Hybridization of atomic orbitals , Molecular Orbital theory

To determine the molecular geometry use the following sequence of steps:

draw Lewis structure → count electron-domain around the central atom → find electron-domain geometry based on position of atoms → determine molecular geometry

The steps for determining the hybridization of the central atom in a molecule are:

draw Lewis Structure of the molecule → count the number of electron-domains on the central atom → determine the hybridization state of the central atom

Bond order is a measure of the strength of a bond and can be determined using a molecular orbital diagram. **Bond order = ½ (# Bonding Electrons – # Antibonding Electrons)**

hybridization

1. Complete the following table. Notice the similarity between the number of electron groups around the central atom and the number of hybrid orbitals.

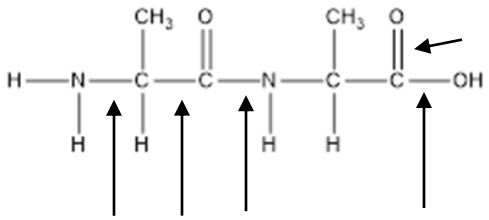
	Chemical Formula	Lewis Structure	# of electron groups around central atom	# of hybrid orbitals	Hybridization around the central atom
a	CO_2				
b	HCN				
c	BCl_3				
d	SO_2				
e	CH_3Cl				
f	NH_3				
g	ClO_3^-				
h	SCl_2				
i	IF_4^+				
j	ClF_3				
k	IF_2^-				
l	XeF_2				
m	BrF_5				
n	XeF_4				

2. Consider the reaction

$$BF_3 + NH_3 \longrightarrow F_3B{-}NH_3$$

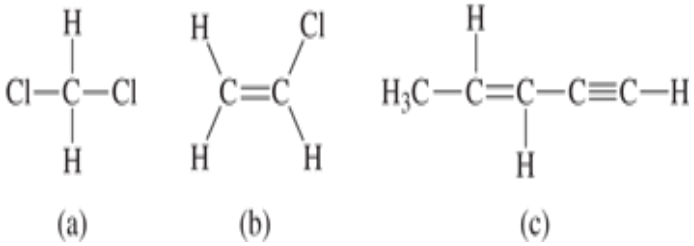
Describe the changes in hybridization (if any) of the B and N atoms as a result of this reaction.

3. Consider the partial Lewis structure shown below (lone pair not shown).
   i. Complete the Lewis structure by adding the lone pairs.
   ii. Describe the bonding scheme for the indicated bonds.
   iii. Indicate the hybridization of each central atom.



4. Sulfur forms oxides, oxoanions, and halides. What is the hybridization of the central S atom in SO_2, SO_3, SO_3^{2-}, SCl_4, SCl_6 and S_2Cl_2.

5. How many sigma bonds and pi bonds are there in each of the following molecules?



(a) (b) (c)

MO Theory

6. Which of the following molecules/ions will not exist according to the MO theory.

   Be_2, Li_2^+, He_2^+, H_2^+

7. Describe the bonding scheme in the C_2^{2-} ion in terms of molecular orbital theory. Compare the bond order in C_2^{2-} with that in C_2.

8. Based on the MO theory, how would you explain the paramagnetic behavior the O_2 molecule.

9. Explain why the bond order of N_2 is greater than that of N_2^+, but the bond order of O_2 is less than that of O_2^+.

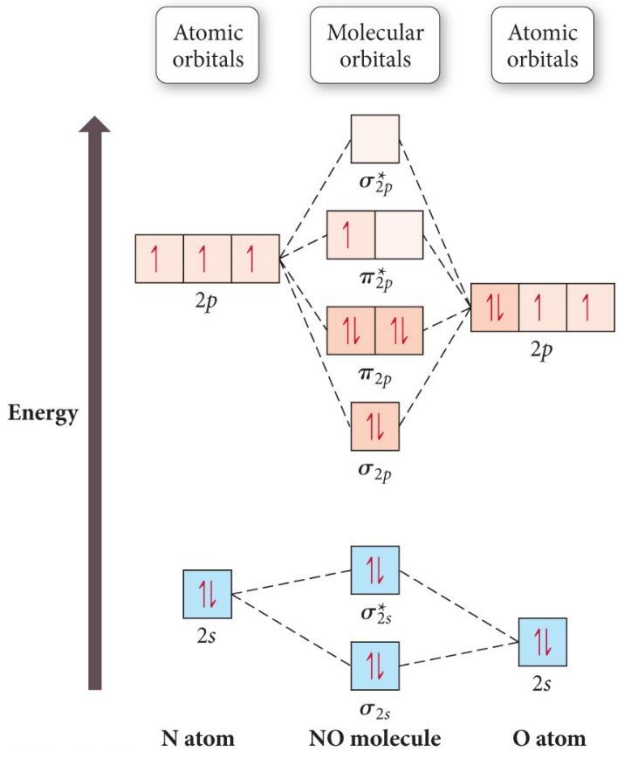
10. Use molecular orbital theory to compare the relative stabilities of F_2 and F_2^+.

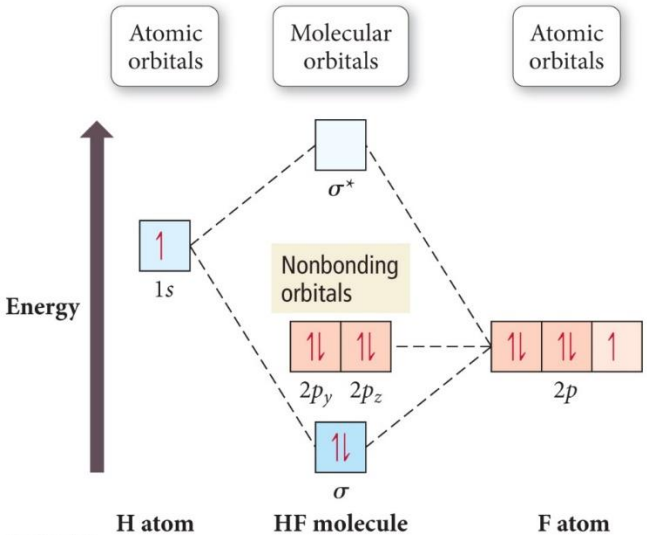
11. Indicate their magnetic properties (that is, diamagnetic or paramagnetic) of the following
Li_2, O_2^{2-}, B_2^{2-}, Ne_2^{2+} C_2^{2+}

12a. Which of the following molecules will show the highest bond order.
Ne_2^{4+}, Ne_2^{2+}, Ne_2

12b. Show the MO diagram for the heteronuclear diatomic molecules NO and HF.

Ans :




Atomic orbitals
Molecular orbitals
Atomic orbitals
Energy
σ*
1s
Nonbonding orbitals
2p_y 2p_z
2p
σ
H atom
HF molecule
F atom